

CA1 Hippocampal Pyramidal Neuron — Synaptic Input Parameters

Biological reference table for NEURON simulation. Excitatory & inhibitory afferents organized by input source, subcellular targeting, conductance, and firing statistics.

All values from primary literature · Dr. Cavarretta Lab

INPUT / AFFERENT SOURCE	SYNAPSE TYPE	NEUROTRAN SMITTER & RECEPTORS	SUBCELLULAR DISTRIBUTION	PEAK CONDUCTANCE (GMAX)	SYNAPSES ONTO ONE CA1 PC	PRESYNAPTIC FIRING FREQUENCY	PRIMARY CITATIONS
► Excitatory Inputs — Glutamatergic Afferents							
CA3 Schaffer Collaterals <i>Hippocampal CA3 → CA1</i>	Excitatory	Glutamate AMPA / NMDA	Stratum radiatum and stratum oriens. CA3c innervates SR; CA3a innervates SO. Proximal–distal apical and basal dendrites. <i>Megias et al. 2001; Bezaire & Soltesz 2013</i>	AMPA: 2.56 ± 0.44 nS (proximal SR) AMPA: 7.96 ± 1.68 nS (distal SR) ~3× distance-dependent scaling NMDA: ~52–71 pA; does not scale with distance	13,059 – 28,697 synapses (ipsilateral + contralateral CA3) 93% target pyramidal cells	0.5–20 Hz baseline <i>Bursts up to 100 Hz during ripples</i>	Andrásfalvy & Magee (2001) J Neurosci PMID:11717348 Nicholson et al. (2006) Neuron PMID:16675397 Bezaire & Soltesz (2013) Hippocampus PMID:23674373 Megias et al. (2001) Neuroscience PMID:11226691
Entorhinal Cortex Layer III <i>Temporoammonic (TA) pathway</i>	Excitatory	Glutamate AMPA / NMDA	Stratum lacunosum-moleculare (distal apical tuft). Both medial and lateral EC project via TA pathway. <i>Megias et al. 2001; Bezaire & Soltesz 2013</i>	AMPA expression lowest in SLM relative to SR — below CA3 SR levels <i>Perforated synapses ~40%. No direct nS value obtainable from papers cited.</i>	< 1,742 synapses (SLM shared with nucleus reuniens)	<i>Insufficient quantitative data available for full parameterization</i>	Nicholson et al. (2006) Neuron PMID:16675397 Bezaire & Soltesz (2013) Hippocampus PMID:23674373 Megias et al. (2001) Neuroscience PMID:11226691
Nucleus Reuniens <i>Thalamic midline — mPFC ↔ RE ↔ CA1</i>	Excitatory	Glutamate AMPA (monosynaptic) Polysynaptic GABA-A via NGF cells	Stratum lacunosum-moleculare. VGLUT2+ terminals restricted to SLM; sparse SR extension. <i>Goswamee et al. 2021</i>	EPSC: -86.8 ± 8.95 pA (at –65 mV holding potential) <i>Largely subthreshold in most CA1 PCs.</i>	<i>Insufficient quantitative data available for full parameterization</i>	Theta: ~5 Hz Gamma bursts: ~40 Hz	Goswamee et al. (2021) Front Synaptic Neurosci PMID:34712120 Bezaire & Soltesz (2013) Hippocampus PMID:23674373
CA2 <i>Hippocampal CA2 → CA1</i>	Excitatory	Glutamate AMPA / NMDA	<i>Insufficient quantitative data available for full parameterization</i>	<i>Insufficient quantitative data available for full parameterization</i>	<i>Insufficient quantitative data available for full parameterization</i>	<i>Insufficient quantitative data available for full parameterization</i>	<i>No papers identified with sufficient quantitative data for CA2→CA1 parameters</i>
► Inhibitory Inputs — Local GABAergic Interneurons							
PV+ Basket Cells <i>Parvalbumin-expressing; fast-spiking</i>	Inhibitory	GABA GABA-A (fast kinetics)	Soma and proximal dendrites (stratum pyramidale). ~48% somatic, ~45% proximal dendritic contacts.	0.95 ± 0.29 nS (range 0.52–1.16 nS) <i>Rise: 4.6 ± 3.2 ms; Decay τ: 32.4 ± 18.0 ms. GABA-A mediated.</i>	183 synapses from ~17 PV+ basket cells 11 ± 0.6 terminals per connection	ISI within theta: 45 ± 25 ms (~22 Hz) <i>Descending theta phase; strongly increases during</i>	Buhl et al. (1995) Eur J Neurosci PMID:8528474 Foldy et al. (2010) Nat Neurosci PMID:20676104 Klausberger et al. (2003) Nature PMID:12594513

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			<i>Klausberger & Somogyi 2008; Buhl et al. 1995</i>		<i>5.8 ± 0.7 terminals on soma</i>	<i>ripples. Paired-pulse ratio: 0.68 ± 0.05</i>	Bezaire & Soltesz (2013) Hippocampus PMID:23674373
CCK+ Basket Cells <i>Cholecystokinin-ex- pressing; CB1R+</i>	Inhibitory	GABA GABA-A (asynchronous)	Soma and proximal dendrites. Terminals extend further onto apical dendrites than PV+ basket cells. <i>Foldy et al. 2010</i>	<i>Insufficient quantitative data available for full parameterization</i>	106 synapses from ~13 CCK+ basket cells 8.3 ± 0.8 terminals per connection 2.3 ± 0.8 somatic terminals	Fires at theta phase when CA1 pyramidal cells begin firing <i>Modulated by endocannabinoids via CB1 receptors (DSI)</i>	Foldy et al. (2010) Nat Neurosci PMID:20676104 Klausberger & Somogyi (2008) Science PMID:18599766 Bezaire & Soltesz (2013) Hippocampus PMID:23674373
Bistratified Cells <i>PV+, SOM+, NPY+; dendrite-targeting</i>	Inhibitory	GABA GABA-A	Basal and oblique dendrites co-aligned with CA3 input (stratum radiatum and stratum oriens). Avoids SLM and soma. <i>Klausberger et al. 2004</i>	Intermediate, variable (electrotonic heterogeneity) <i>Rise: 2.0 ± 0.2 ms; Decay: 16.1 ± 1.1 ms. Paired-pulse ratio: 1.09 ± 0.18 (facilitation)</i>	104 synapses from ~10 bistratified cells 10–11 synaptic contacts per connection 76% target small/medium dendritic shafts	5.9 ± 3.0 Hz (theta) 42.8 ± 4.4 Hz (ripples) <i>Theta trough; tightest gamma coupling; onset 64 ms before ripple start</i>	Klausberger et al. (2004) Nat Neurosci PMID:14634650 Maccaferri et al. (2000) J Physiol PMID:10747186 Bezaire & Soltesz (2013) Hippocampus PMID:23674373
O-LM Interneurons <i>Oriens-lacunosum moleculare; SOM+</i>	Inhibitory	GABA GABA-A	Distal apical tuft in stratum lacunosum-moleculare co-aligned with entorhinal input. 93% boutons in SLM, 7% in stratum oriens. <i>Maccaferri et al. 2000</i>	~0.37 nS (converted from 26 ± 10 pA at -70 mV) <i>Smallest amplitude of all interneuron types. Rise: 6.2 ± 0.6 ms; Decay: 20.8 ± 1.7 ms</i>	77 synapses from ~8 O-LM cells 10 synapses per connection 72 on distal apical dendrites, 5 on basal	ISI within theta: 76 ± 19 ms (~13 Hz) <i>Theta trough; SILENT during sharp-wave ripples. Paired-pulse ratio: 0.93 ± 0.04</i>	Maccaferri et al. (2000) J Physiol PMID:10747186 Klausberger et al. (2003) Nature PMID:12594513 Bezaire & Soltesz (2013) Hippocampus PMID:23674373
Ivy Cells <i>nNOS+, NPY+; most abundant interneuron</i>	Inhibitory	GABA + NPY + NO GABA-A (slow) Volume transmission	Proximal basal and oblique dendrites co-aligned with CA3 input (stratum oriens and stratum radiatum). Avoids SLM. <i>Fuentealba et al. 2008</i>	~0.09–0.14 nS (from IPSC 6.1–9.9 pA at -60 mV) <i>Rise: 2.6–3.0 ms; Half-width: 15.8–17.8 ms. 78% of boutons use volume transmission.</i>	422 synapses from ~42 ivy cells 10 classical synapses per connection 81% target small dendritic shafts. 37% of all CA1 GABAergic neurons.	0.7 ± 0.7 Hz (theta) 0.8 ± 1.1 Hz (ripples) <i>Theta trough (30.7° ± 63.1°); silent during >90% of ripple episodes.</i>	Fuentealba et al. (2008) Neuron PMID:18367092 Bezaire & Soltesz (2013) Hippocampus PMID:23674373 Klausberger & Somogyi (2008) Science PMID:18599766
Neurogliaform Cells <i>α-actinin-2+, COUP-TFII+, reelin+</i>	Inhibitory	GABA GABA-A (slow) + GABA-B Volume transmission	Distal apical tuft in stratum lacunosum-moleculare. Dense local axonal plexus; axon crosses hippocampal fissure. <i>Fuentealba et al. 2010</i>	<i>Insufficient quantitative data available for full parameterization</i>	140 synapses from ~14 neurogliaform cells 10 classical synapses per connection Majority of boutons via volume transmission	2–10 Hz (theta) 0.4–4.2 Hz (ripples) <i>n=2 labeled cells — small sample. Fires just after theta peak (192–199°),</i>	Fuentealba et al. (2010) J Neurosci PMID:20130170 Bezaire & Soltesz (2013) Hippocampus PMID:23674373 Armstrong et al. (2012) Front Neural Circ PMID:22623913

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						<i>opposite phase to ivy cells.</i>	
Septal GABAergic Inputs <i>Medial septum → CA1; theta pacemaker</i>	Inhibitory	GABA / ACh GABA-A	<i>Insufficient quantitative data available for full parameterization</i>	<i>Insufficient quantitative data available for full parameterization</i>	Minor contribution to direct pyramidal cell inhibition <i>Mainly targets interneurons, not pyramidal cells directly</i>	<i>Insufficient quantitative data available for full parameterization</i>	Bezaire & Soltesz (2013) Hippocampus PMID:23674373

Full References

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